

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1543

Capote 138 kV
April 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1543. The Interconnection Customer proposes a 186.3 MW battery energy storage facility interconnecting at the Capote 138 kV substation in the West Virginia area. The request was received on August 05, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 442 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,202,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Chadwick-S Point 138 kV Line
Contingency: Tristate-Valemount 345 kV line
Loading: 129.1% (Rating: 265 MVA)
DFAX: 63.0%
Pre-project loading: 84.9%
2. Facility: S Pt-Capote 138 kV line
Contingency: Valemount-Tristate 345 kV
Loading: 129.1% (Rating: 264 MVA)
DFAX: 63.0%
Pre-project loading: 84.6%
3. Facility: Gable Street-Capote 138 kV
Contingency: Valemount-Tristate 345 kV
Loading: 110.6% (Rating: 260 MVA)
DFAX: 37.0%
Pre-project loading: 84.1%
4. Facility: Eagle Mountain-Sukukna 138 kV
Contingency: Chadwick-S Pt 138 kV
Loading: 103.6% (Rating: 178 MVA)
DFAX: 74.0%
Pre-project loading: 26.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chadwick-South Pt 138 kV	Rebuild line with higher-capacity conductor	\$6,184,000	72% (\$4,443,000)
S Pt-Capote 138 kV line	Rebuild line with higher-capacity conductor	\$13,782,000	72% (\$9,901,000)
Capote-Gable Street Substation 138 kV	Rebuild line with higher-capacity conductor	\$7,782,000	60% (\$4,668,000)
Eagle Mountain-Sukukna 138 kV	Reconductor line	\$3,190,000	100% (\$3,190,000)
	Total allocated to project		\$22,202,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.